Name:

CS403/503 Programming Languages
Spring 2025
Assignment #2

1. Problem 7 of Chapter 5 on Page 229

Answer:

Static scoping yields x = 5; dynamic scoping yields x = 10.

2. Problem 10 of Chapter 5 on Page 231

Answer:

Point 1:

a. Definition 1
b. Definition 2
c. Definition 2
d. Definition 2

Point 2:

a. Definition 1
b. Definition 2
c. Definition 3
d. Definition 3
e. Definition 3

Point 3:

a. Definition 1
b. Definition 2
c. Definition 2
d. Definition 2

Point 4:

a. Definition 1
b. Definition 1
c. Definition 1

3. Problem 12 (b-f) of Chapter 5 on Page 232

Answer: Assuming dynamic scoping is being used,

b) a, x, w declared in sub3; y, z declared in sub1
c) a, y, z declared in sub1; x, w declared in sub3; b declared in sub2
d) a, y, z declared in sub1; x, w declared in sub3
e) a, b, z declared in sub2; x, w declared in sub3; y declared in sub1
f) a, y, z declared in sub1; b declared in sub2; x, w declared in sub3

4. Review Question 37 of Chapter 6 on Page 296

Answer:

Reference counters are used in the eager approach to garbage reclamation. For every cell, reference counters keep track of the number of pointers that are pointing at it at any given time. A decrement operation takes place within the reference pointer when a pointer is disconnected from a cell, and the value is examined to see if it is zero. In the event that it is 0, the cell is marked as garbage and added back to the list of available space. The mark-sweep operation, another name for the lazy approach to trash reclamation, first permits garbage to build up while allocating storage cells and releasing pointers from the cells as required. Then, with the aid of an additional indicator bit present in every cell, the mark-sweep procedure begins. When this bit is set, the mark sweep operation designates reachable cells as non-garbage and allocates the pointers of every cell in the heap. After that, it moves on to the last stage, sometimes referred to as the sweep phase, where every cell that wasn't designated as "not garbage" is added back to the list of available space.

5. Describe the tombstone and lock-and-key methods of avoiding dangling pointers.

Answer:

- Tombstone method
    - Safety
        - Tombstone stops a pointer from ever pointing to a deallocated variable and safely deallocates pointers by assigning them incorrect values
        - Because they can precisely show if a pointer is allocated while using indirection, tombstones are intrinsically safe
    - Implementation cost
        - Tombstones need to be assigned; they are never deallocated
            - Negatively impacts space cost
        - Accessing variable through tombstone requires an extra machine cycle
            - Negatively impact time cost
- Lock and key method
    - Safety
        - Although Lock and Key is generally safe, it compromises certain security for speed and functionality

        - For instance, a particular heap dynamic variable can be referenced by any number of pointers because only the lock value needs to be modified
        - Each copy will therefore keep its address, but access will be refused as the key and lock value do not match
    - Implementation cost
        - Memory must be allocated for the lock value in both the header cell and key value cell
            - Minimally negative impacts space cost
        - Key and lock values must be compared
            - Negatively impacts time cost